Intro to System Dynamics

EGR 345

GVSU

Outline

- what is this class about?
  - what is a dynamic system?
  - what is feedback control?
  - why do these things matter?
- who am I?
- who are you?
- my teaching philosophy and the importance of this course?

What is a dynamic system?

- a system whose transient response matters
    - when the input changes, it does not immediately go to the new output
        - something happens before it settles at the new output state
- a system whose input/output relationship involves differential equations

What is feedback control?

Controls engineering in west Michigan

- what is a controls engineering in west Michigan?
    - what does that have to do with this class?

Is this class really important?

Will you learn more…

- will you learn more if you believe this course is important?
- will you learn more if you understand my teaching philosophy?

Are there dynamic systems in the real world?

Is feedback control really important?

Mugshots and Intro to Dr. Krauss

Syllabus

My Teaching Philosophy

Who cares?

- should you care about my teaching philosophy?
- will you learn more if you understand my teaching philosophy?

Strengths and Weaknesses of the Padnos College of Engineering (PCE)

Strength: PCE is known for its co-op program

- why is this a strength?
- how might this also be a weakness?

Strength: PCE is good at projects and hands on stuff

- how is this a strength?
  - this is really important and a key part of our “brand”
- how might this also be a weakness?

Strength: PCE is good at prototyping

- strengths:
    - a prototype is worth a thousand pictures
    - low-fidelity prototyping is a great idea
    - others?
- how might this also be a weakness?
    - what is wrong with trial and error?
    - JCI armrest story

Quote from Advisory Board Member

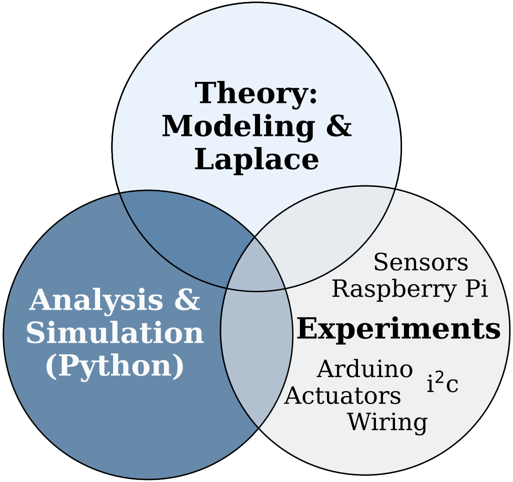
- 20-30% of his engineers are GVSU alumni

“All good engineers figure out how to apply theory to real-world problems. GVSU engineers do that better than most.”

Key Ideas

- GVSU graduates engineers who are good at real-world application and hands-on stuff
- our alumni are not as strong in theory, analysis, and simulation (in my opinion)

This is a challenging class

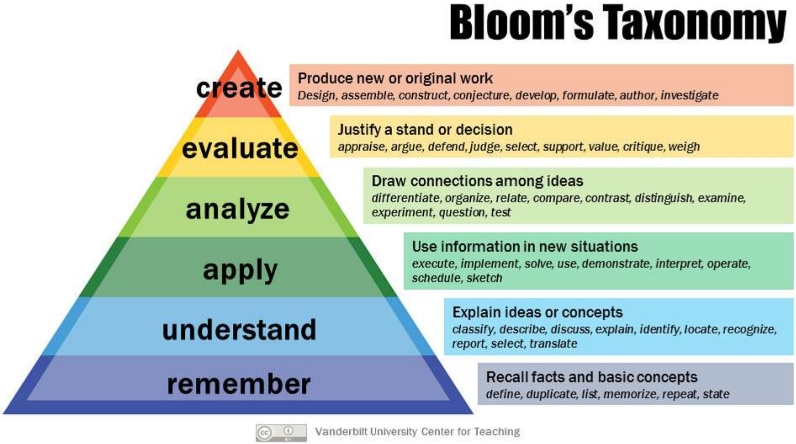


Course Goals

- I want to push you to grow in modeling, simulation, and analysis-driven design
- why do I want to do this?

My Teaching Philosophy

Bloom's Taxonomy



- memorization is necessary, but pretty low-level learning

Active Learning and the Socratic Method

- I believe strongly in active learning and the Socratic method
  - passively listening to a lecture is not a great way to learn
  - my colleagues believe in active learning as well, but handle it differently

Why not just teach like everyone else?

- you have been trained by the typical engineering teaching style:
    - professor reads and explains the slides
    - professor does a few examples
    - some questions are entertained
- why don’t I just do that?
- origin story of my teaching style
    - watching learning happen as a tutor at MTU

Two Quotes from my LIFT Evaluations

- “At first the ‘active learning’ seemed like a stretch to be effective, but once the ball got rolling, it was actually my preferred method of teaching. . . .**This is the only professor I’ve had to go through the ‘active learning’ process, and I have to say, I really wish that other classes were taught this way.**”
- “Honestly, **the active learning style that (Dr.) Krauss preaches is very productive and made my comprehension better and workload feel lighter**. I feel like I put much less work into this class, not because it was too easy but because (Dr.) Krauss really made sure that we understood the topics before the end of each class and it directly translated to my ability to complete homework in a timely fashion. This was refreshing because the homework was very reflective of the work we did together in class.”

What is this class really about?

I think this class is about transfer functions

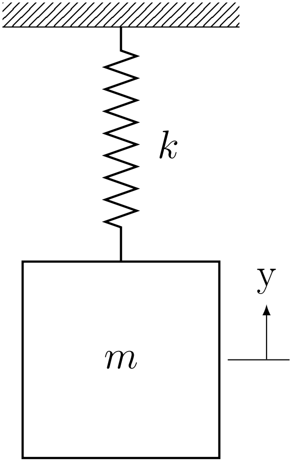
$$In(s) \longrightarrow \boxed{G(s)} \longrightarrow Out(s)$$

- transfer functions are awesome and powerful

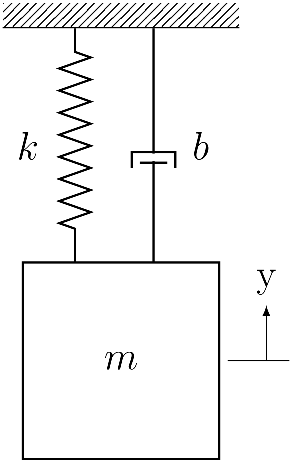
Trying to make things less abstract: Mass/Spring/Damper Systems

- this is (now) a course about mass/spring/damper systems
  - the building blocks of mechanical dynamic systems
  - using transfer functions and Laplace to analyze MSD systems and design feedback control systems

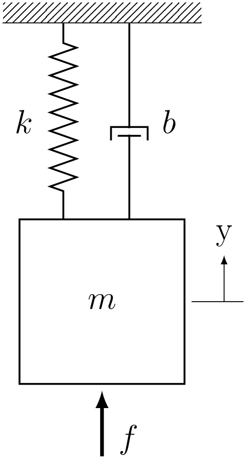
Step 1: Mass and Spring



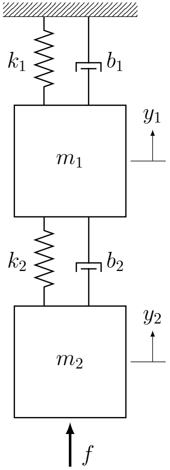
Step 2: Mass/Spring/Damper (no input)



Step 3: Mass/Spring/Damper with external input



Step 4: Multiple Masses



Step 5 and beyond

- vibration suppression control
- other applications